MODULE 1 · LECTURE 27

Automatic Weeding and Harvesting Systems Using AI

Why weeding and harvesting are being automated, how camera-guided hoes, in-row robots, spot sprayers and lasers remove weeds, why timing and position accuracy decide crop safety, and how fruit-harvesting robots detect, reach, grip and detach produce — with worked examples, a robotic-hoe program and the performance record of harvesting robots.

Learning objectives

After this lecture you should be able to:

1. Explain why weeding and selective harvesting are being automated.
2. Distinguish inter-row and intra-row weeding and describe camera-guided, robotic, spot-spraying and laser weeding.
3. Calculate the timing and position accuracy an AI weeder needs at different forward speeds.
4. Interpret the trade-off between weeds removed and crop plants damaged.
5. Describe the steps and components of a fruit-harvesting robot and how its performance is measured.
6. Discuss the progress, limits and future directions of weeding and harvesting robots, including Indian conditions.

Scope: Lecture 24 explained autonomous machines and field robots, and Lecture 25 AI detection of weeds and spot spraying. This lecture covers the robots that remove weeds and harvest produce. Sorting and grading of the harvested produce follow in Lecture 28.

1. Why automate weeding and harvesting?

- **Labour**: weeding and selective harvesting are among the most labour-intensive farm tasks. Selective harvesting of high-value crops such as apples, tomatoes and broccoli is still done mainly by people, making it one of the most expensive tasks, while labour shortage is becoming a limiting factor (Kootstra et al., 2021).
- **Herbicides**: herbicide use has led to herbicide-resistant weeds and environmental contamination, while mechanical weeding can disturb the soil and trigger new flushes of weeds (Andreasen et al., 2022).
- **Specialty crops**: many fruit and vegetable crops have few effective herbicides, because they are small markets for the agrochemical industry; robotic weeders are cheaper to develop and less regulated than new herbicides (Fennimore & Cutulle, 2019).
- **Indian conditions**: weeding and harvesting create sharp labour peaks in the season; on small fields, light robots offered as a hired service are more practical than large machines (Lecture 24).

2. Automatic weeding

Where the weeds are

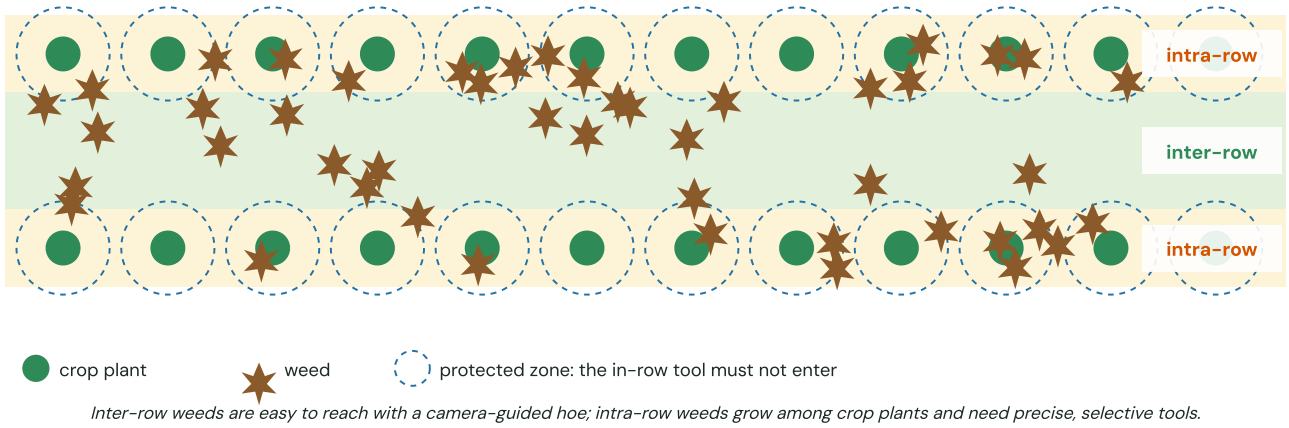


Figure 1. Inter-row and intra-row zones in a row crop. Original diagram.

Inter-row weeds grow between the crop rows, away from crop plants, and can be cut by hoes guided along the rows. **Intra-row** weeds grow in the row among the crop plants; removing them without harming the crop requires the machine to know where each crop plant is — this is where AI is essential.

Method	How it works	Notes
Camera-guided inter-row hoe	A camera follows the crop rows; a hydraulic side-shift keeps the hoe blades centred between them	Robust, reliable and widely available from manufacturers (Machleb et al., 2020); works at higher speed and with better selectivity and efficacy than manual steering (Gerhards et al., 2022)
Robotic in-row (intra-row) weeder	AI detects each crop plant; tools move into the row between plants and out round them	Combines row guidance with in-row hoeing; selectively controls individual weeds within rows (Gerhards et al., 2022)
Spot spraying	Single nozzles spray only detected weeds (Lecture 25)	Large herbicide savings; still uses chemicals
Laser weeding	Deep learning locates weeds; a robotic actuator aims laser energy at them	Treats a small area, leaves the soil undisturbed; small vehicles have limited capacity and laser reflections can harm people and animals; not yet widely practised (Andreasen et al., 2022)

Sensors used with mechanical weeding include cameras, GNSS, laser and ultrasonic systems, each with advantages and disadvantages; fully automated robotic systems still struggle with the ever-changing field environment (Machleb et al., 2020).

Timing: why speed matters

The camera sees a plant a short distance ahead of the tool. Between seeing it and reaching it, the machine must process the image, decide and move the blade at exactly the right moment. Any variation in this time becomes a position error equal to speed × time.

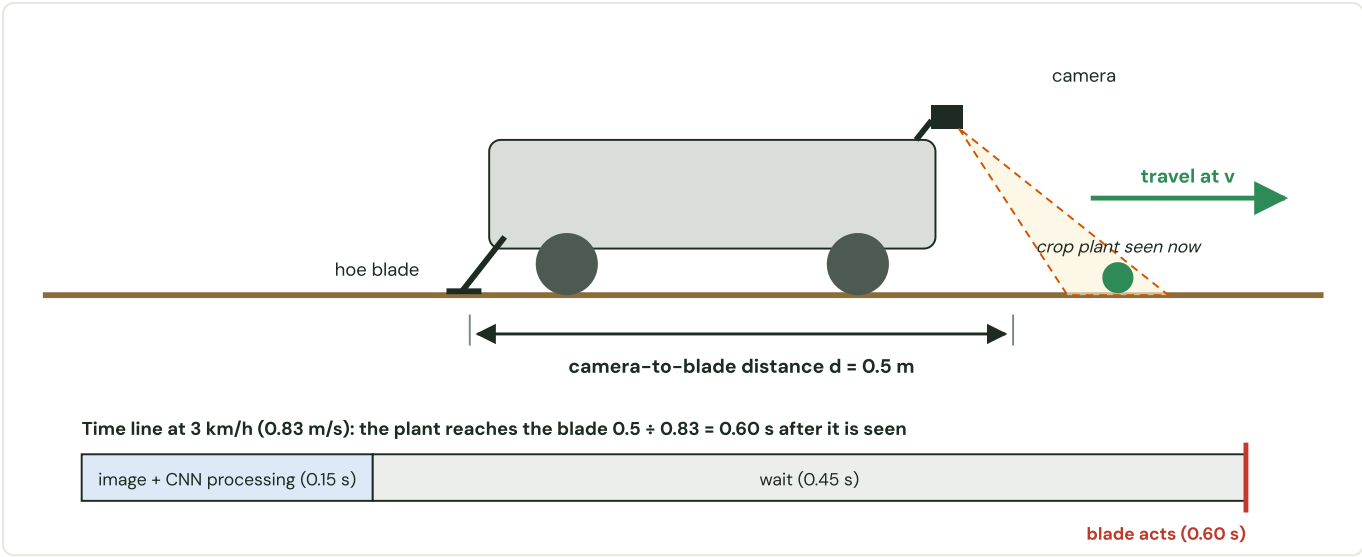


Figure 2. A camera 0.5 m ahead of the hoe blade gives the computer a fixed time window, which shrinks as speed rises. Original diagram.

Worked example 1 — the timing budget of an AI weeder
Camera 0.5 m ahead of the blade. At 3 km/h (0.83 m/s) a plant reaches the blade 0.5 ÷ 0.83 = **0.60 s** after it is seen. If the CNN takes 0.15 s, the controller must fire the blade 0.45 s after the result.
A timing variation (jitter) of 30 ms gives a position error of 0.83 × 0.03 = **2.5 cm**.
At 5 km/h (1.39 m/s): only 0.36 s is available, and the same 30 ms jitter gives **4.2 cm** error.
If the CNN needed 0.40 s at 5 km/h, the camera would have to be at least 0.40 × 1.39 = 0.56 m ahead — at 0.5 m the answer would arrive too late.
Faster weeding needs faster processing, a longer camera-to-tool distance and a larger safety margin round each crop plant.

3. Worked example: an AI-guided in-row hoe

The program simulates one 100 m crop row with a plant every 25 cm and 1,200 weeds. The CNN recognises 98% of crop plants; their estimated positions carry a 1 cm image error plus the timing jitter of Worked example 1. The hoe works the whole row except a protected window round each detected plant. The program tests four window sizes at 2 and 5 km/h.

Simulating an AI-guided in-row hoe in Python (Lecture 8)

```
import random

random.seed(8)
ROW = 100.0                                  # length of crop row, m
crops = [0.25 * i + random.gauss(0, 0.03) for i in range(1, 400)]   # crop every 25 cm
weeds = [random.uniform(0, ROW) for _ in range(1200)]               # weeds anywhere in the row

def run(speed_kmh, radius_cm):
    """The camera finds crop plants; the hoe works the row except a protected window round each."""
    v = speed_kmh / 3.6                      # forward speed, m/s
    windows, damaged = [], 0
    for c in crops:
        if random.random() > 0.98:           # 2% of crop plants not recognised by the CNN
            damaged += 1
            continue
        # position error: image measurement (1 cm) + timing jitter of the actuator (30 ms)
        est = c + random.gauss(0, 0.01) + v * random.gauss(0, 0.03)
        lo, hi = est - radius_cm / 100, est + radius_cm / 100
        windows.append((lo, hi))
        if not lo <= c <= hi:
            damaged += 1                     # blade entered where the crop actually stands
    protected = lambda x: any(lo <= x <= hi for lo, hi in windows)
    removed = sum(1 for w in weeds if not protected(w))
    return 100 * removed / len(weeds), 100 * damaged / len(crops)

print("speed  protected radius  weeds removed  crop plants damaged")
for speed in (2, 5):
    for radius in (2, 4, 6, 8):
        w, d = run(speed, radius)
        print(f"{speed} km/h       {radius} cm            {w:5.1f}%            {d:4.1f}%")

Output:
speed  protected radius  weeds removed  crop plants damaged
2 km/h       2 cm             85.8%            30.8%
2 km/h       4 cm             70.6%             4.8%
2 km/h       6 cm             53.2%             1.5%
2 km/h       8 cm             39.5%             1.8%
5 km/h       2 cm             85.1%            64.9%
5 km/h       4 cm             70.1%            40.9%
5 km/h       6 cm             55.6%            18.3%
5 km/h       8 cm             37.9%             7.0%
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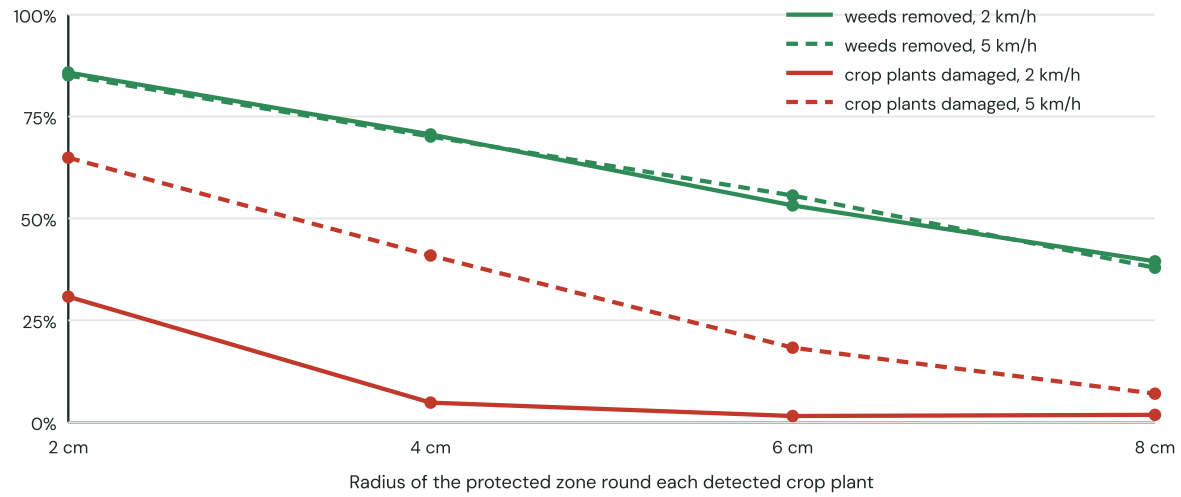


Figure 3. Weeds removed and crop plants damaged against the size of the protected zone, at two speeds. Original chart from the program's output.

- **Small protected zone (2 cm)**: 85–86% of in-row weeds removed, but 31% of crop plants damaged at 2 km/h and 65% at 5 km/h — far too risky.
- **At 2 km/h**, a 6 cm zone cuts damage to 1.5% while removing 53% of in-row weeds. Damage cannot fall below about 2% here, because 2% of crop plants are not recognised at all — detection accuracy sets the floor.
- **At 5 km/h**, the larger timing error means even an 8 cm zone still damages 7% of plants, while the weeds removed fall to 38%: speed costs either crop safety or weed control.
- A protected zone of 8 cm on each side of plants 25 cm apart covers 64% of the row, so many weeds close to crop plants survive; in practice these are left to crop competition, a second pass or another tool.

The program was run and the output shown is its actual output. Plant spacing, error sizes and detection rate are illustrative; the pattern — accuracy and speed set the balance between weed control and crop damage — holds for real in-row weeders.

4. Automatic harvesting

Bulk harvesting of grains, by combine harvesters, is already mechanised and increasingly automated with auto-steering and yield monitors (Lectures 24 and 26). **Selective harvesting** — picking only ripe fruits, one at a time, without damage — is much harder for a robot because of high variation among plants and fruits, incomplete information when fruits are hidden by leaves, and the need for safety around people (Kootstra et al., 2021). Robots are being developed for greenhouses, orchards and open fields.

How a harvesting robot works

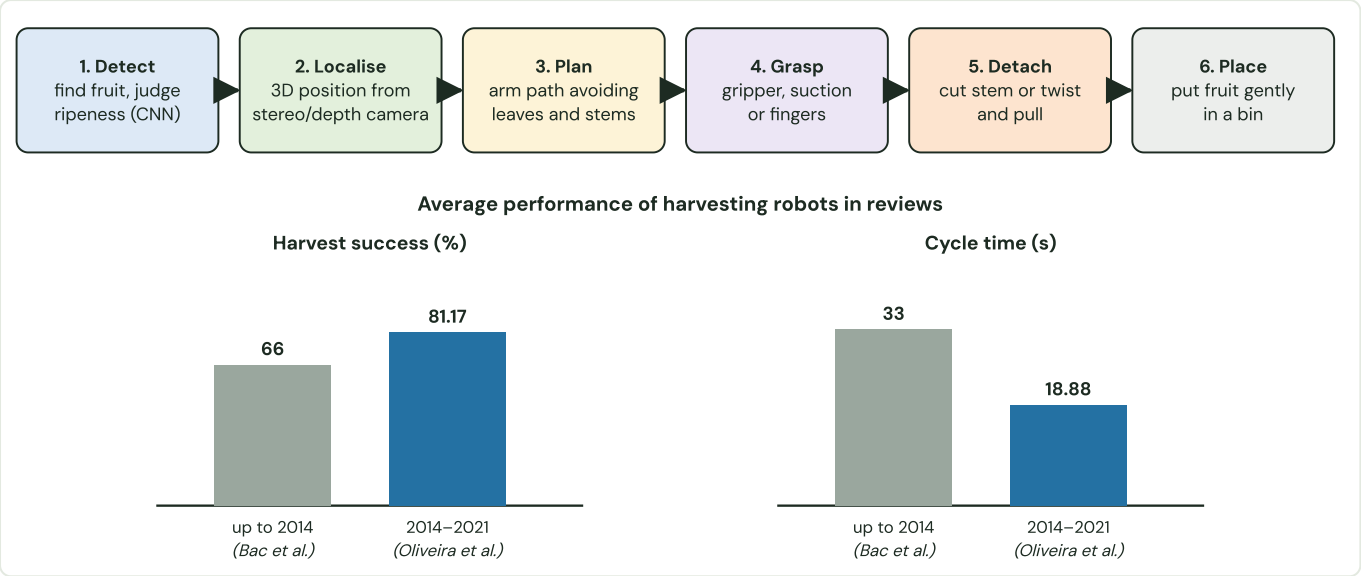


Figure 4. Top: the six steps of selective robotic harvesting. Bottom: average harvest success and cycle time reported in two reviews. Original diagram and chart.

Performance measure	Meaning
Localisation success	Share of fruits correctly found and located
Detachment success	Share of located fruits successfully picked
Harvest success	Share of all ripe fruits harvested
Fruit or peduncle damage	Share of fruits or stems damaged
Cycle time	Average time to pick one fruit

Bac et al. (2014) reviewed 50 harvesting robots for high-value crops developed over three decades. On average, localisation success was 85%, detachment success 75%, harvest success 66%, fruit damage 5%, peduncle damage 45% and cycle time 33 s (the fastest, a kiwi robot, took 1 s). Performance had not improved over the three decades and none of the 50 robots had been commercialised. Reviewing robots from 2014–2021, Oliveira et al. (2021) found an average harvest success of 81.17% (range 50–100%) and cycle time of 18.88 s (2–36.9 s) — about 23% higher success and 43% shorter cycles — but only two of the harvesting robots they listed had reached the commercialisation phase.

Bac et al. (2014) proposed four directions: **simplify the task** (for example, train plants so that fruits hang visible and reachable), **enhance the robot**, **define requirements and measure performance** in the same way, and **consider what is needed for practical use**.

> **Worked example 2 — how many robot arms are needed?**
> With the 2014–2021 average: 3,600 ÷ 18.88 = 191 attempts per hour × 0.8117 success = **≈ 155 fruits per hour per arm**; working 20 h a day → ≈ 3,100 fruits/day.
> With the earlier average: 3,600 ÷ 33 × 0.66 = **72 fruits per hour** → 1,440 fruits/day.
> A greenhouse block needing 12,000 fruits picked per day would need 12,000 ÷ 3,100 ≈ 3.9 → **4 arms** at the newer average, but 12,000 ÷ 1,440 ≈ 8.3 → **9 arms** at the older one.
> The 19–34% of fruits the robots miss must still be picked by people, so robots and workers usually share the work.

The daily target and 20-hour working day are illustrative assumptions; success rates and cycle times are the review averages quoted above.

5. Benefits and challenges

Benefits	Challenges
Less dependence on scarce seasonal labour (Kootstra et al., 2021)	Field variation and hidden fruits or plants make perception hard
Fewer herbicides; tools for crops with few registered herbicides (Fennimore & Cutulle, 2019)	Trade-off between speed, weed removal and crop damage (Worked example 1, Figure 3)
Laser and mechanical tools leave no chemical residue (Andreasen et al., 2022)	Low capacity of small robots; safety hazards such as laser reflections
Robots can work long hours and record what they did	Harvest success and cycle times still limit output; few systems have been commercialised (Bac et al., 2014; Oliveira et al., 2021)
Camera-guided hoes are already reliable and widely available (Machleb et al., 2020)	High cost for small farms; need for local service, training and suitable crop layouts

6. Key takeaways

- Labour shortage, herbicide resistance and few herbicides for specialty crops drive automatic weeding; labour cost drives harvesting robots.
- Inter-row weeding is solved by camera-guided hoes; intra-row weeding needs AI to locate each crop plant.
- In-row hoes, spot sprayers and lasers act only where weeds are; lasers leave soil undisturbed but have low capacity and safety risks.
- Position error = speed × timing error: faster weeding needs faster processing and larger safety margins.
- The protected-zone size trades weeds removed against crop damage; detection accuracy sets the minimum damage.
- Harvesting robots detect, localise, plan, grasp, detach and place each fruit; success and cycle time are the key measures.
- Average harvest success rose from 66% (to 2014) to about 81% (2014–2021) and cycle time fell from 33 s to about 19 s, but few robots are commercial.

7. Examination questions

Short-answer questions (2 marks)

Q1. Distinguish inter-row and intra-row weeds.

Inter-row weeds grow between crop rows; intra-row weeds grow within the row among crop plants.

Q2. Why is intra-row weeding difficult to automate?

The tool must work close to crop plants, so it must know the exact position of each plant.

Q3. What is a camera-guided hoe?

An inter-row hoe whose position is kept between the rows by a camera and a hydraulic side-shift.

Q4. Give two advantages and one risk of laser weeding.

Targets small areas, leaves soil undisturbed, no chemicals; laser reflections can harm people and animals.

Q5. Calculate the time a weeder has at 3 km/h if the camera is 0.5 m ahead of the blade.

0.5 ÷ 0.83 = 0.60 s.

Q6. Why does weeding speed affect crop damage?

Position error equals speed × timing error, so the same jitter causes larger errors at higher speed.

Q7. Distinguish bulk and selective harvesting.

Bulk harvesting gathers a whole crop at once (e.g. combines); selective harvesting picks individual ripe fruits.

Q8. Name the six steps of a harvesting robot.

Detect, localise, plan, grasp, detach and place.

Q9. What is cycle time?

The average time a robot takes to harvest one fruit.

Q10. State the average harvest success found by Bac et al. (2014).

66%, with an average cycle time of 33 s.

Long-answer questions (5–10 marks) — answer outlines

Q1. Discuss the need for automation of weeding and harvesting.

Labour, herbicide resistance, specialty crops; Indian labour peaks and small fields.

Q2. Describe the methods of automatic weeding with their merits and limits.

Figure 1; methods table; camera-guided hoes, in-row robots, spot spraying, lasers; sensors.

Q3. Explain the timing and accuracy requirements of an AI weeder with a numerical example.

Figure 2; worked example 1; speed × jitter; processing time and camera distance.

Q4. Explain the trade-offs in robotic in-row weeding using simulation results.

Program; Figure 3; protected zone, speed, detection floor; practical implications.

Q5. Describe how a fruit-harvesting robot works and how its performance is measured.

Figure 4 steps; measures table; Bac et al. averages; Oliveira et al. progress.

Q6. Discuss the progress and challenges of harvesting robots.

Review results; four directions; worked example 2; benefits and challenges table.

8. Glossary

Term	Meaning
Bulk harvesting	Harvesting a whole crop at once, as by a combine
Camera-guided hoe	Inter-row hoe kept between rows by camera and side-shift
Cycle time	Average time to harvest one fruit
Detachment	Separating the fruit from the plant
End-effector	Tool at the end of a robot arm: gripper, suction cup, cutter
Harvest success	Share of ripe fruits harvested
Inter-row	Space between crop rows
Intra-row	Space within the crop row, among crop plants
Jitter	Random variation in timing
Laser weeding	Killing weeds with targeted laser energy
Localisation	Finding the 3D position of a fruit or plant
Peduncle	Stalk attaching a fruit to the plant
Protected zone	Area round a crop plant that weeding tools must avoid
Selective harvesting	Picking individual ripe fruits
Side-shift	Sideways movement of an implement to follow rows
Spot spraying	Spraying only detected weeds with single nozzles

References

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Fennimore, S. A., & Cutulle, M. (2019). Robotic weeders can improve weed control options for specialty crops. *Pest Management Science, 75*(7), 1767–1774. https://doi.org/10.1002/ps.5337

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Oliveira, L. F. P., Moreira, A. P., & Silva, M. F. (2021). Advances in agriculture robotics: A state-of-the-art review and challenges ahead. *Robotics, 10*(2), 52. https://doi.org/10.3390/robotics10020052

All diagrams, charts, examples and programs are original. Machine values and the simulated crop row are illustrative; every calculation and the program output were checked by running them.

Practice MCQs

Q1. Weeds growing within the crop row among the crop plants are called

(a) inter-row weeds
(b) intra-row weeds
(c) border weeds
(d) aquatic weeds

Q2. A camera-guided inter-row hoe keeps its blades between the rows by

(a) GPS only
(b) a hydraulic side-shift controlled from camera images
(c) manual steering
(d) a laser

Q3. Which weeding method leaves the soil undisturbed but needs care because reflections can harm people?

(a) Inter-row hoeing
(b) Laser weeding
(c) Ploughing
(d) Flame weeding by hand

Q4. With the camera 0.5 m ahead of the blade at 3 km/h, a plant reaches the blade after about

(a) 0.06 s
(b) 0.6 s
(c) 6 s
(d) 1.8 s

Q5. A 30 ms timing error at 5 km/h causes a position error of about

(a) 0.4 cm
(b) 4.2 cm
(c) 15 cm
(d) 42 cm

Q6. In the robotic hoe simulation, increasing the protected zone round crop plants

(a) removes more weeds and damages more crops
(b) removes fewer weeds and damages fewer crops
(c) has no effect
(d) only changes speed

Q7. In the simulation, crop damage never fell below about 2% because

(a) the blade was blunt
(b) 2% of crop plants were not recognised by the CNN
(c) the speed was too low
(d) weeds were too many

Q8. Picking only ripe fruits one at a time is called

(a) bulk harvesting
(b) selective harvesting
(c) threshing
(d) winnowing

Q9. The correct order of steps for a harvesting robot is

(a) grasp, detect, place, detach
(b) detect, localise, plan, grasp, detach, place
(c) place, plan, detect, grasp
(d) detach, detect, grasp, place

Q10. Bac et al. (2014) found an average harvest success of

(a) 33%
(b) 66%
(c) 85%
(d) 99%

Q11. Bac et al. (2014) reported that of the 50 harvesting robots reviewed

(a) all were commercial
(b) none was commercialised
(c) half were commercial
(d) most used lasers

Q12. At an average cycle time of 18.88 s and 81% success, one arm harvests about

(a) 19 fruits/h
(b) 155 fruits/h
(c) 1,000 fruits/h
(d) 3,600 fruits/h

Answers

Q1 — (b) intra-row weeds

Q2 — (b) a hydraulic side-shift controlled from camera images

Q3 — (b) Laser weeding
Q4 — (b) 0.6 s
Q5 — (b) 4.2 cm
Q6 — (b) removes fewer weeds and damages fewer crops
Q7 — (b) 2% of crop plants were not recognised by the CNN
Q8 — (b) selective harvesting
Q9 — (b) detect, localise, plan, grasp, detach, place
Q10 — (b) 66%
Q11 — (b) none was commercialised
Q12 — (b) 155 fruits/h